Level 3



CLOUDFLIGHT CODING CONTEST

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LEVEL 3

THE PROFESSOR IS OUT OF TOWN

A sticky note on the fume hood:

Need 8 flasks of C19H21Cl3N6O8S2 by Monday. Budget's tight. Don't blow anything up.

– F.

Still dazed after glancing at the new shop prices, you look for the professor's notebook in a pile of receipts.

You stare at one of the pages trying to make sense of the intricate web of reactions for fifteen minutes before realizing you were holding it upside down.

You are going to need to think this through.

Buying fresh compounds for every batch is how labs go broke.

The cheapest factory wins.

Now it’s your turn to run the factory.

- You are given a list of molecules that can be bought
- You are given a list of possible reactions
  - Some molecules can only be bought
  - Some molecules can only be produced by reactions
  - Some molecules can be both bought and produced

- You are given a list of target molecules and quantities, for each:
  - Start with an empty inventory
  - **Produce at least the given quantity of the target molecule**
  - Reactions can be performed any amount of times
  - Reactions must only use molecules that are in your inventory or can be bought



This level is an optimization level.

- Your score is the total cost of the bought molecules
- A lower score is better

Input

Name	Description	Example
M	Number of buyable molecules	3 2 -> H2 1 -> O2 3 -> C 4 2 H2, 1 O2 -> 2 H2O 1 C, 1 O2 -> 1 CO2 6 CO2, 6 H2O -> 1 C6H12O6, 6 O2 1 C6H12O6 -> 2 C2H5OH, 2 CO2 2 2 C6H12O6 4 C2H5OH
Cost to molecule (Repeated M times)	The cost and a molecule separated by an arrow “ -> ”	
Empty line		
R	Number of available reactions	
Reaction (Repeated R times)	Two molecule list separated by an arrow “ -> ”	
Empty line		
N	Number of cases	
Target molecule (Repeated N times)	Target molecule count and molecule name separated by space “ ”	

Output

Name	Description	Example
Reactions paragraph (Repeated N times)	List of reactions, one per line	1 C, 1 O2 -> 1 CO2 2 H2, 1 O2 -> 2 H2O 1 C, 1 O2 -> 1 CO2 ... 1 C, 1 O2 -> 1 CO2 2 H2, 1 O2 -> 2 H2O 1 C, 1 O2 -> 1 CO2 ...
	Mandatory empty line between paragraphs	



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